Geometry-Constrained Rock–TBM Spatial Interaction Graphs for Chainage-Resolved Diagnostic Analysis of Excavation Responses

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Abstract

Existing monitoring-based TBM response models preserve temporal dependence, but they do not explicitly represent the spatial relations between geological units and TBM components. This study proposes a chainage-referenced rock–TBM spatial interaction graph model for diagnostic analysis of excavation responses. TSP-derived rock voxels and parameterized TBM surface components are formalised as heterogeneous spatial entities, and candidate rock–machine relations are constrained by active-zone membership, distance decay, surface-normal compatibility, and excavation state. Based on the resulting graph sequence, component–chainage spatial interaction descriptors are derived by aggregating a fixed-reference low-velocity anomaly score over geometry-weighted candidate relations. Instead of claiming direct contact force, calibrated jamming risk, or broad forecasting superiority, the descriptors are evaluated through their consistency with persistence residuals of TBM monitoring responses. Case studies in BSLL DyK1017+205 and SJLS Dyk1252+411 show that the proposed graph representation organises rock–machine relations into interpretable component–chainage patterns and provides diagnostic information that monitoring curves alone do not explicitly encode. The contribution is therefore a geometry-constrained spatial entity-relation representation and residual-consistency diagnostic workflow for rock–TBM interaction, rather than a new deep prediction model.

Keywords: spatial representation; heterogeneous graph; rock–machine interaction; geometry-constrained relation graph; component–chainage descriptor; persistence residual; TBM excavation; diagnostic analysis
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1 Introduction

Tunnel boring machines (TBMs) operate through a moving contact zone in which geological conditions, machine geometry, and operating state jointly shape excavation responses. Abnormal thrust, torque, penetration, advance rate, or shield-pressure behaviour may indicate that the local rock–machine interaction state has changed. Monitoring curves record the temporal expression of these responses, but they do not by themselves identify which geological units are near the TBM, which machine components are implicated, or how these spatial relations evolve along chainage.

This limitation is not only a forecasting problem. Traditional monitoring models represent recent observations as a temporal sequence, $u_{t-K+1:t} \rightarrow r_{t+h}$, and can exploit the strong inertia

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of TBM responses. They do not, however, answer spatial relation questions: which TSP voxel is close to the cutterhead or shield at a given step, whether the proximity is geometrically compatible with the local surface normal, or how the candidate rock–machine relation set changes as excavation advances [1, 4, 5, 14]. Rock voxels, TBM surface components, and candidate interaction relations therefore form a heterogeneous spatial entity-relation problem rather than a plain time-series problem.

This study addresses that representation problem by constructing chainage-referenced rock–TBM spatial interaction graphs. TSP-derived rock voxels and parameterized TBM surface samples are placed in a unified local chainage coordinate system. Candidate rock–machine edges are then screened by active-zone membership, distance decay, surface-normal compatibility, and excavation state. The resulting graph at each excavation step explicitly represents the spatially plausible relation set $G_t = (V_t^r \cup V_t^m, E_t^{rr} \cup E_t^{mm} \cup E_t^{rm})$, where E_t^{rm} is not an arbitrary learned connectivity pattern but a geometry-constrained candidate relation set.

The paper does not claim that a deep forecasting model is the main solution. The available effective samples and current response-modelling results do not support broad forecasting superiority over persistence. Instead, the central method is an explicit spatial interaction descriptor. A geometry-compatible edge weight summarises the plausibility of each candidate rock–machine relation, and a restrained TSP-derived geological anomaly score summarises local unfavourability. Aggregating these quantities over TBM components and chainage produces component–chainage descriptors that can be compared with persistence residuals, i.e., the part of the monitoring response not already explained by temporal inertia.

The contributions are threefold. First, the study proposes a chainage-referenced rock–TBM spatial entity model that places TSP voxels, TBM surface components, and monitoring responses in a common excavation coordinate system. Second, it defines a geometry-constrained rock–machine candidate relation graph using active-zone filtering, distance decay, surface-normal compatibility, and excavation-state constraints. Third, it derives component–chainage spatial interaction descriptors from geometry-weighted TSP anomaly summaries and evaluates their consistency with persistence residuals of TBM responses. These descriptors are diagnostic summaries of plausible spatial relations; they are not contact-force measurements, jamming probabilities, or calibrated operational risk estimates.

2 Related work

2.1 TBM response prediction and jamming-related analysis

Existing TBM prediction and warning studies have shown that tabular models and sequence models can be effective for forecasting penetration, thrust, torque, or jamming-related responses [1, 4, 14]. Sequence models are effective for learning temporal dependence in monitoring records [5], but they usually treat excavation as an ordered signal sequence rather than as a spatially structured interaction process. As a consequence, existing studies usually represent geological and machine variables as feature vectors or sequences, rather than explicit spatial relations between geological entities and TBM components. This limitation matters when the objective is not only response prediction, but also spatially consistent interpretation of interaction dynamics.

2.2 Spatial representation of geological conditions in tunnel excavation

Tunnel seismic prediction (TSP) and advance geological prediction provide spatially distributed descriptions of the geological field ahead of or around excavation [6]. TSP-derived geological information can be transformed into chainage-referenced rock voxels that preserve spatial support and local geological context. Voxelised geological models and three-dimensional geological representation have been used to describe subsurface heterogeneity along tunnel alignments. However, this representation still needs to be linked to machine geometry: a geological voxel field alone does not specify which rock units are candidate interaction partners for which TBM surface components at a given excavation step. Bridging geological spatial representation and machine surface geometry is therefore a prerequisite for spatially explicit rock–machine interaction modelling.

2.3 Graph-based spatial relation modelling and spatial interaction descriptors

Graph concepts are useful for this problem because they provide a formal way to represent irregular spatial objects, contiguity, and interaction. In GIScience, graphs have long been used to express spatial neighbourhoods and interaction relations [12]. Recent IJGIS and IJDE studies further show that graph-based representations can organise spatial units, neighbourhood structure, and urban network morphology [2, 8]. The important point for the present study is not that a graph must be used as a deep learning architecture, but that it can define an explicit spatial entity–relation data model.

Spatial interaction models also emphasise that relations are not arbitrary: they are shaped by distance decay, compatibility, direction, and spatial support. This logic is directly relevant to rock–TBM analysis, where a candidate rock–machine relation should depend on active-zone membership, surface proximity, surface-normal compatibility, and excavation state. A graph therefore provides a disciplined representation of plausible candidate relations before any predictive model is considered.

Graph neural networks and heterogeneous graph networks remain relevant as possible extensions for learning on such structured data [11, 15], and graph-based forecasting models have been successful in other spatiotemporal domains [3, 7, 13, 16]. However, the present paper does not position the graph primarily as a black-box forecasting backbone. It uses the graph to derive auditable component–chainage spatial interaction descriptors from geometry-constrained rock–machine candidate relations.

3 Methodology

This section formalizes rock–TBM interaction as a chainage-referenced spatial entity-relation graph. The method has four steps: defining rock and machine spatial entities, constructing geometry-constrained candidate relations, aggregating component–chainage spatial interaction descriptors, and evaluating descriptor consistency with persistence residuals of monitoring responses.

3.1 Chainage-referenced rock–TBM spatial entity model

Let excavation advance be discretized into chainage-indexed steps $t = 1, 2, \ldots, T$. The monitoring vector at step t is

$$u_t = [\mathrm{AdvanceRate}_t,\ \mathrm{RPM}_t,\ \mathrm{Torque}_t,\ \mathrm{Thrust}_t,\ \mathrm{Penetration}_t,\ \mathrm{ShieldPressure}_t],$$

and the response variables used for residual diagnostics are denoted $r_t^{(k)}$, where k indexes variables such as torque, thrust, penetration, advance rate, or shield pressure.

The geological input is a TSP-derived voxel field $D_{\mathrm{geo}} = \{(c_i, g_i)\}$, where $c_i = (x_i, y_i, z_i)$ is the coordinate of rock voxel i and g_i is its geological attribute vector. The machine input is a parameterized TBM surface $M_{\mathrm{TBM}} = \{p_j(0), n_j, \rho_j\}$, where $p_j(0)$ is the initial surface node coordinate, n_j is the local surface normal, and ρ_j is the machine component label. The current implementation partitions the machine surface into cutterhead, front shield, middle shield, and tail shield components.

Because TSP coordinates and TBM monitoring mileage are commonly recorded in different local frames, all entities are mapped to a common local chainage coordinate. A simple alignment is

$$x_{\text{local}}(t) = \text{ShieldMileage}(t) - \text{ShieldMileage}_{\text{start}},$$

after which TBM surface nodes are advanced along the excavation direction d:

$$p_j(t+1) = p_j(t) + \Delta x\, d.$$

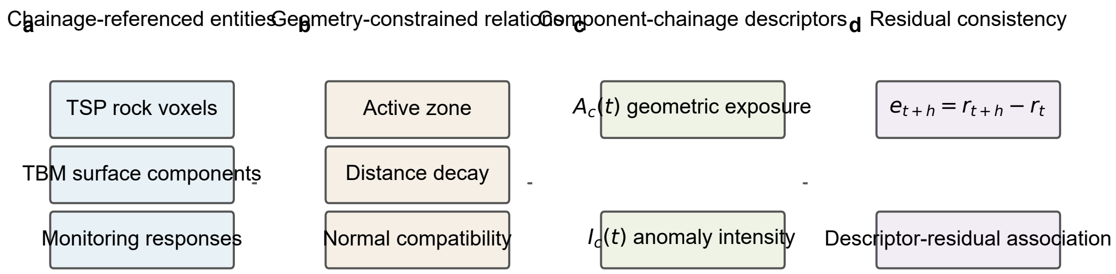

Figure 1: Overview of the proposed chainage-referenced rock–TBM spatial interaction graph workflow. The main manuscript focuses on spatial entities, geometry-constrained candidate relations, component–chainage descriptors, and response-residual consistency. The descriptors are not contact pressure, contact force, jamming probability, or calibrated risk.

3.2 Geometry-constrained rock–machine candidate relation graph

At each excavation step, a graph snapshot is defined as

$$G_t = \left(V_t^r \cup V_t^m,\ E_t^{rr} \cup E_t^{mm} \cup E_t^{rm}\right),$$

where V_t^r and V_t^m are rock and TBM surface node sets, E_t^{rr} denotes rock–rock voxel adjacency, E_t^{mm} denotes TBM surface adjacency, and E_t^{rm} denotes candidate rock–machine relations.

Rock node features are

$$x_i^r(t) = [c_i, g_i, s_i(t)],$$

where $s_i(t)$ records whether voxel i remains active and eligible for interaction at step t. TBM node features are

$$x_j^m(t) = [p_j(t), n_j(t), \rho_j].$$

The intra-rock edge set E_t^{rr} is constructed from voxel neighbourhoods, and E_t^{mm} is constructed from local neighbourhoods on the parameterized TBM surface. The key graph-design decision is the construction of E_t^{rm}.

A candidate edge (i, j) is included in E_t^{rm} only when four conditions are satisfied:

$$\begin{cases} d_{ij}(t) \le \tau_{\text{edge}}, \\ \kappa_{ij}(t) \ge \eta_{\min}, \\ c_i \in \Omega_t, \\ s_i(t) = 1, \end{cases}$$

where

$$d_{ij}(t) = \|c_i - p_j(t)\|$$

is the distance between rock voxel i and TBM surface node j, and

$$\kappa_{ij}(t) = \max\left(0, \frac{n_j(t)^\top (c_i - p_j(t))}{d_{ij}(t) + \epsilon}\right)$$

is the signed-normal compatibility. The active zone Ω_t restricts candidate relations to the excavation-relevant rock volume, while $s_i(t)$ prevents already excavated voxels from repeatedly contributing to later graph snapshots. It is defined as the union of the face look-ahead zone and the shield-adjacent zone:

$$\Omega_t = \Omega_t^{\text{face}} \cup \Omega_t^{\text{shield}}.$$

Here Ω_t^{face} contains voxels ahead of the cutterhead within the look-ahead distance used for face interaction screening, and Ω_t^{shield} contains voxels within the shield-side annular buffer used for shield interaction screening. This definition separates face-contact plausibility from shield-side proximity while keeping both parts within one active rock volume for graph construction.

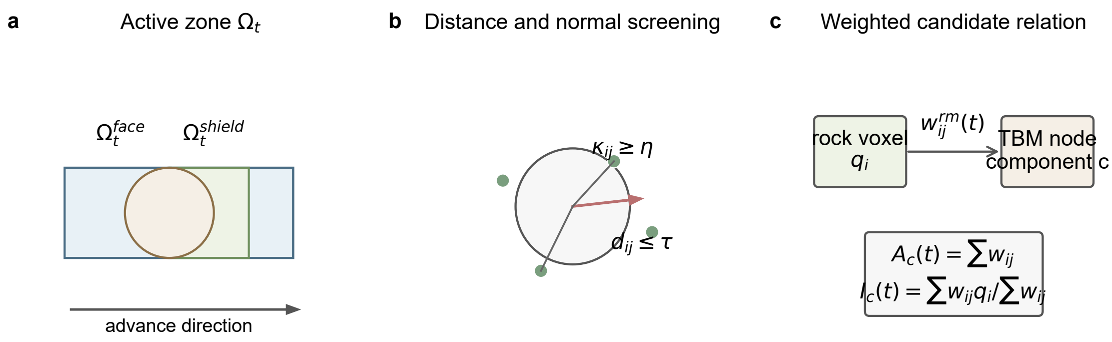

Figure 2: Geometry-constrained rock–machine candidate relation construction. Candidate relations are controlled by active-zone membership, distance decay, surface-normal compatibility, and excavation state. The construction defines spatially plausible relations; it does not recover measured contact.

For each candidate relation, the geometric interaction weight is defined as

$$w_{ij}^{rm}(t) = \exp\left(-\frac{d_{ij}(t)}{\tau_{\text{edge}}}\right) \kappa_{ij}(t).$$

This weight is the explicit form of geometry-compatible interaction intensity used by the main descriptors. It increases when a rock voxel is closer to the TBM surface and more compatible with the local surface normal. It remains a geometric descriptor, not a contact force or pressure.

3.3 Component–chainage spatial interaction descriptors

The geological contribution of each rock voxel is represented by a restrained TSP-derived anomaly score

$$q_i = \phi(g_i).$$

In the main formulation, q_i is a bounded low-velocity anomaly score derived from the training active-zone distribution of P-wave velocity:

$$q_i = \text{clip}\left(\frac{Q_{95}^{\text{train}}(Vp) - Vp_i}{Q_{95}^{\text{train}}(Vp) - Q_5^{\text{train}}(Vp) + \epsilon}, 0, 1\right).$$

This fixed training-reference scaling avoids per-chainage min–max normalisation and keeps descriptor values comparable along chainage. Lower V_p values therefore correspond to higher geological anomaly scores under a common reference distribution. This single-attribute definition is intentionally transparent and reproducible across the BSLL and SJLS cases. Multi-attribute variants can be examined as sensitivity checks, but they are not required for the main descriptor.

For each TBM component c, let V_c^m denote the set of TBM surface nodes whose component label is c. Two component–chainage descriptors are then defined. The pure geometric exposure is

$$A_c(t) = \sum_{j \in V_c^m} \sum_{i:(i,j) \in E_t^{rm}} w_{ij}^{rm}(t),$$

where $A_c(t)$ is a total geometric exposure measure. It is affected by component surface extent, surface-node sampling density, candidate-edge count, and active-zone geometry. It is therefore used mainly to describe total candidate-relation exposure; when direct cross-component comparison is required, a node-normalised or area-normalised exposure can be used. The geology-weighted spatial interaction intensity is

$$I_c(t) = \frac{\sum_{j \in V_c^m} \sum_{i:(i,j) \in E_t^{rm}} w_{ij}^{rm}(t) q_i}{\sum_{j \in V_c^m} \sum_{i:(i,j) \in E_t^{rm}} w_{ij}^{rm}(t) + \epsilon}.$$

$A_c(t)$ describes how strongly component c is geometrically exposed to the screened candidate rock–machine relation set, whereas $I_c(t)$ describes the weighted geological anomaly intensity around that component. These descriptors are defined for cutterhead, front shield, middle shield, and tail shield at each chainage step, producing a component-by-chainage diagnostic map.

3.4 Response-residual consistency analysis

Persistence is a strong baseline for short-horizon TBM monitoring responses. The proposed graph descriptors therefore do not attempt to replace temporal inertia. Instead, they are evaluated against persistence residuals:

$$e_{t+h}^{(k)} = r_{t+h}^{(k)} - r_t^{(k)},$$

where h is the diagnostic horizon and k indexes a response variable. The diagnostic question is whether component–chainage descriptors co-vary with the remaining response change after persistence has been removed:

$$I_c(t) \to e_{t+h}^{(k)}.$$

This analysis can be reported using Pearson or Spearman correlations, overlap between high-descriptor and high-absolute-residual intervals, and component-wise comparisons. The result is interpreted as descriptor–response residual co-variation within the evaluated chainage interval, not as causal attribution.

4 Case studies and experimental design

4.1 Case roles and data sources

The experiments use two real tunnel cases with complementary evidential roles. The BSLL tunnel case starts from DyK1017+205 and is evaluated under two diagnostic horizons: h=1 and h=3. It provides a compact case for testing whether component–chainage descriptors can be constructed and related to response residuals over short target intervals. The SJLS tunnel case starts from Dyk1252+411 and provides the clearest external TSP contrast, with a longer test interval for descriptor–residual association.

For each excavation step, the monitoring vector contains advance rate, cutterhead RPM, torque, thrust, penetration, and shield pressure. The diagnostic responses are advance rate, torque, thrust, penetration, and shield pressure. The geological input is a TSP-derived voxel field. For BSLL, the project TSP voxel field contains P-wave velocity, S-wave velocity, density, dynamic Young's modulus, velocity ratio, and Poisson's ratio around the tunnel alignment. For SJLS, external P-wave and S-wave velocity profiles at Dyk1252+411 are preprocessed into the same project TSP schema and paired with mileage-aligned monitoring records.

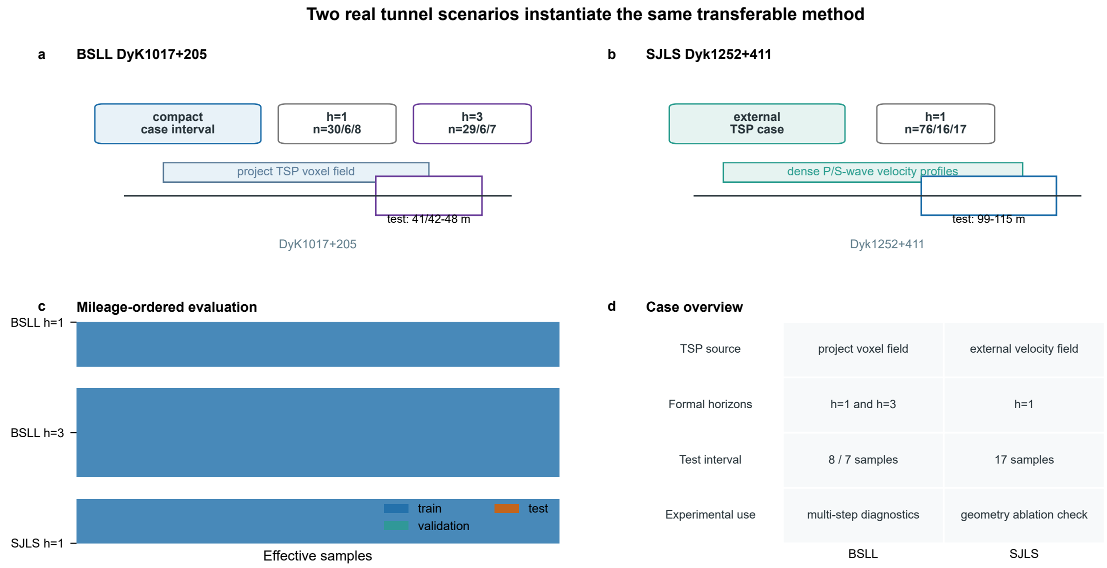

Figure 3: Real tunnel scenarios used to instantiate the common descriptor framework. BSLL DyK1017+205 provides compact h=1 and h=3 cases, while SJLS Dyk1252+411 provides the clearest external TSP descriptor evidence.

4.2 Samples and graph-construction settings

All cases use mileage-ordered partitioning. Descriptor values are computed from the last observed graph snapshot of each target sample, and the last observed response at step t is used to define the persistence residual for a target response at $t + h$. The formal case runs are summarised in Table 1.

Table 1: Formal case runs used in the descriptor experiments. Effective samples are chainage-indexed residual samples after applying the diagnostic horizon.

Case	Tunnel	Start chainage	Horizon	Train/Val./Test samples
BSLL h=1	BSLL	DyK1017+205	1	30 / 6 / 8
BSLL h=3	BSLL	DyK1017+205	3	29 / 6 / 7
SJLS h=1	SJLS	Dyk1252+411	1	76 / 16 / 17

The BSLL one-step run has target chainages 41–48 m in the test interval. The BSLL three-step run has target chainages 42–48 m. The SJLS run has target chainages 99–115 m. The graph construction uses $\tau_{\text{edge}} = 2.0$ m, $\eta_{\min} = 0.3$, and $\tau_{\text{zone}} = 5.0$ m in the main setting. The TBM surface is discretised into cutterhead, front-shield, middle-shield, and tail-shield surface nodes at a resolution of 0.5 m.

TSP preprocessing follows a common voxel schema. Coordinates are converted to a local excavation coordinate by subtracting the minimum longitudinal coordinate, so the local X axis is aligned with advance direction. Each voxel stores $[V_p, V_s, E, V_p/V_s, \nu, \rho]$, and these attributes are standardised using statistics fitted on the training active-zone margin. This multi-attribute schema is retained to keep the BSLL and SJLS inputs in a common voxel format. However, the main spatial interaction descriptor uses only the cross-case consistent P-wave velocity attribute. The geological anomaly score is therefore a $[0, 1]$ low-velocity anomaly score derived from the training active-zone reference distribution of V_p; derived or case-specific attributes such as V_s, modulus, velocity ratio, Poisson's ratio, and density are not used in the main descriptor calculation.

4.3 Diagnostic evaluation metrics

The primary outputs are component–chainage descriptors. For each component c, $A_c(t)$ reports the sum of geometry weights over candidate relations, and $I_c(t)$ reports the geometry-weighted mean TSP anomaly score. The response consistency target is the persistence residual $e_{t+h}^{(k)} = r_{t+h}^{(k)} - r_t^{(k)}$ for each response variable k.

Descriptor–residual association is reported with Pearson and Spearman correlations and with overlap between high-descriptor and high-absolute-residual intervals. A threshold sensitivity analysis varies τ_{edge} and $\eta_{\min}$ to check whether the descriptor–residual pattern is tied to one specific graph-construction setting. The results are interpreted as chainage-local diagnostic co-variation, not as causal evidence or calibrated operational risk.

5 Results

5.1 Constructed rock–machine interaction graph sequences

The first result is the constructed spatial relation object itself. For each case, TSP-derived rock voxels and parameterized TBM surface nodes are mapped to the same chainage-referenced coordinate system, and the candidate relation set E_t^{rm} is rebuilt at each excavation step. The graph is therefore not a conceptual diagram: it is an explicit sequence of screened rock–machine relations controlled by active-zone membership, distance threshold, surface-normal compatibility, and excavation state.

Figure 2 illustrates the main construction rule. The important diagnostic consequence is that each candidate relation is traceable to a rock voxel, a TBM surface component, a chainage step, a distance, a normal compatibility value, and a geometry weight $w_{ij}^{rm}(t)$. This traceability is what monitoring-only curves do not provide.

Table 2 summarises the constructed graph objects in the test intervals. The BSLL runs use the same tunnel section with different diagnostic horizons and therefore have the same graph size per step, whereas SJLS uses a smaller parameterized TBM surface but a comparable active-zone rock voxel field. The candidate-edge counts confirm that the subsequent descriptors are computed from explicit rock–machine relation sets rather than from conceptual links.

Table 2: Graph construction statistics in the test intervals. Edge shares are computed from candidate rock–machine relations in the main graph setting.

Case	Rock nodes	TBM nodes	Candidate edges	Cutterhead share	Shield share
BSLL h=1	3558	1352	14620	0.145	0.855
BSLL h=3	3558	1352	14620	0.145	0.855
SJLS h=1	3558	352	3834	0.159	0.841

5.2 Component–chainage spatial interaction patterns

The main output of the revised framework is the component–chainage descriptor map built from $A_c(t)$ and $I_c(t)$. In each case, the cutterhead, front-shield, middle-shield, and tail-shield values are computed across the test chainage interval as geometry-weighted TSP anomaly summaries over candidate rock–machine relations. They are not contact pressure, contact force, jamming probability, or calibrated risk.

Figure 4 summarises the descriptor evidence chain. Panel (a) shows component–chainage $I_c(t)$ patterns in all three formal cases. Panel (b) reports representative fixed descriptor–residual pairs for subsequent sensitivity analysis. Panel (c) confirms that the result is based on explicitly constructed candidate rock–machine relations.

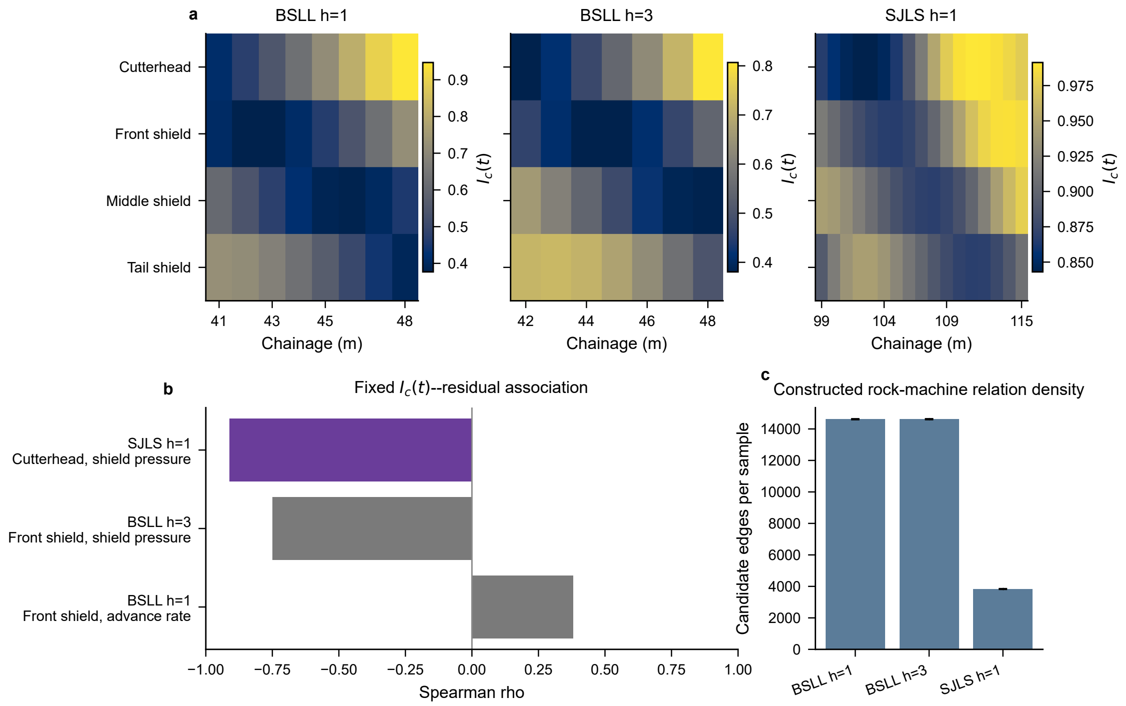

Figure 4: Component–chainage spatial interaction descriptor evidence. Panel (a) shows $I_c(t)$ values for BSLL h=1, BSLL h=3, and SJLS h=1, aggregated from geometry-weighted TSP anomaly scores over candidate rock–machine relations. Colour scales are case-specific and should not be compared across panels. Panel (b) reports representative fixed $I_c(t)$–persistence-residual pairs used in the threshold sensitivity analysis; the complete association results are reported in Table 3 and Figure 5. Panel (c) reports the mean number of constructed candidate relations per sample. The descriptors are not measured contact pressure or risk probability.

5.3 Association with persistence residuals

The revised evaluation target is response-residual consistency rather than absolute response forecasting. For each response variable k, the persistence residual is

$$e_{t+h}^{(k)} = r_{t+h}^{(k)} - r_t^{(k)}.$$

The descriptor evidence reports the association between $I_c(t)$ and $e_{t+h}^{(k)}$ across the evaluated chainage interval. Table 3 gives the full component–response Spearman matrix for $I_c(t)$. These correlations are exploratory and unadjusted for multiple testing; they are used to diagnose whether the spatial descriptor contains response-consistent structure, not to claim a calibrated operational rule. The associated p-values are reported only as descriptive indicators of within-case association strength. They are not used as confirmatory hypothesis tests because the matrices are exploratory, the samples are limited, and no multiplicity adjustment is applied.

The full matrix is treated as the primary diagnostic result. The SJLS case shows the clearest descriptor–residual structure, including a negative cutterhead–shield-pressure association ($\rho = -0.912$) and a strong positive tail-shield–shield-pressure association ($\rho = 0.775$). These patterns are interpreted as case-specific diagnostic co-variation within the evaluated chainage interval rather than universal component–response rules. BSLL h=3 shows a suggestive front-shield $I_c(t)$–shield-pressure residual association ($\rho = -0.750$, $p = 0.052$), while BSLL h=1 remains weaker ($\rho = 0.381$, $p = 0.352$ for the front-shield advance-rate pair).

Table 3: Full unadjusted component–response Spearman matrix for $I_c(t)$–persistence-residual association. Correlations are computed over the test chainage interval and are interpreted as exploratory descriptor diagnostics.

Case	Component	Adv. rate	Torque	Thrust	Penetr.	Shield press.
BSLL h=1	Cutterhead	0.238	-0.286	0.048	-0.095	0.095
BSLL h=1	Front shield	0.381	-0.429	0.190	0.095	-0.095
BSLL h=1	Middle shield	-0.048	0.262	0.452	-0.167	0.143
BSLL h=1	Tail shield	-0.238	0.286	-0.048	0.095	-0.095
BSLL h=3	Cutterhead	-0.214	-0.714	-0.036	-0.214	-0.536
BSLL h=3	Front shield	0.286	-0.179	0.143	0.250	-0.750
BSLL h=3	Middle shield	0.214	0.714	0.036	0.214	0.536
BSLL h=3	Tail shield	0.179	0.750	0.107	0.143	0.571
SJLS h=1	Cutterhead	0.860	0.446	-0.669	0.669	-0.912
SJLS h=1	Front shield	0.640	0.152	-0.350	0.475	-0.745
SJLS h=1	Middle shield	-0.140	-0.431	0.397	-0.223	-0.108
SJLS h=1	Tail shield	-0.843	-0.314	0.740	-0.804	0.775

Figure 5 visualises the same matrix to make the component–response structure easier to inspect. Values are exploratory and unadjusted for multiple testing; the figure is used to diagnose case-specific descriptor–residual co-variation rather than confirm universal component–response rules. Together, Table 3 and Figure 5 provide the full result set, whereas Figure 4(b) only highlights representative fixed pairs for threshold-sensitivity checking.

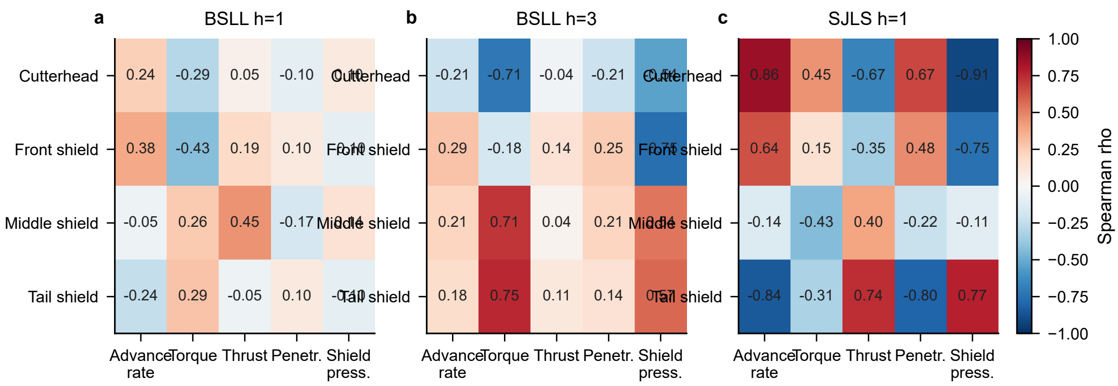

Figure 5: Full component–response Spearman correlation matrix between $I_c(t)$ and persistence residuals. Values are exploratory and unadjusted for multiple testing; they are used to diagnose case-specific descriptor–residual co-variation rather than confirm universal component–response rules.

5.4 Sensitivity to graph-construction thresholds

Figure 6 tests whether the fixed $I_c(t)$–residual pairs are preserved when the edge-distance threshold τ_{edge} and normal-compatibility threshold η_{min} are varied. The descriptor–response pairs used in Figure 4(b) and Figure 6 were selected from the main-setting component–response matrix and then held unchanged across all threshold combinations. They are representative diagnostic pairs for threshold-sensitivity checking rather than confirmatory component–response rules. For each case, the same pair is kept fixed over all τ_{edge} and η_{min} combinations; the sensitivity plot therefore evaluates threshold stability rather than searching for a new best pair.

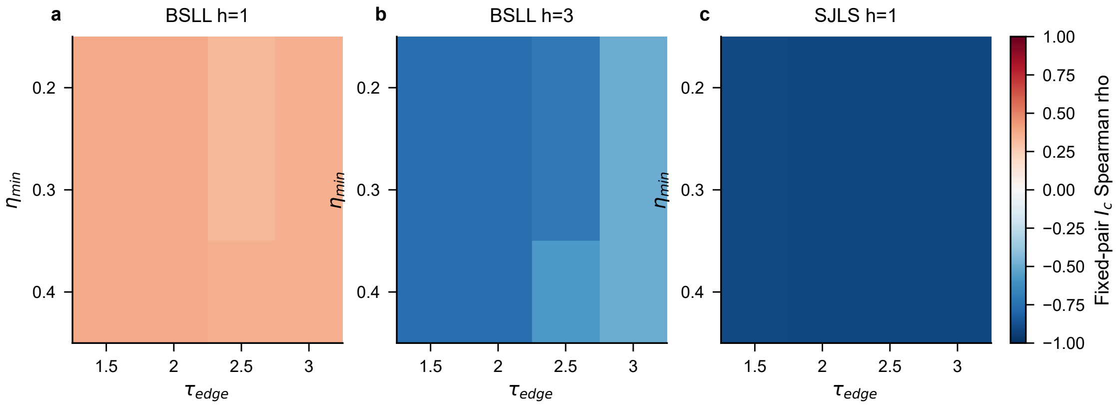

Figure 6: Sensitivity of fixed $I_c(t)$–residual Spearman pairs to graph-construction thresholds. Each panel varies τ_{edge} and η_{min} for one formal case while holding the descriptor–response pair fixed. The plot evaluates descriptor stability, not prediction accuracy.

5.5 Case-specific interpretation and remaining checks

The difference between SJLS and BSLL should be interpreted in relation to sample length and geological–response contrast. SJLS contains a longer test interval and a clearer external TSP velocity contrast, which makes descriptor–residual co-variation more visible. BSLL h=1 and h=3 are compact intervals with 8 and 7 test samples, respectively, and the response residuals may be more strongly affected by operational control and local construction variability. The weaker BSLL evidence therefore does not invalidate the graph construction, but it limits the strength of case-specific residual-consistency claims.

The current evidence supports a bounded descriptor claim. Alternative definitions of q_i, spatial detrending, and additional tunnel cases remain necessary before stronger diagnostic claims are made.

6 Discussion

6.1 Why graph representation is necessary

The revised manuscript separates two questions that were previously too closely coupled: whether a deep graph sequence model improves forecasting, and whether rock–TBM interaction requires a spatial entity-relation representation. The current evidence does not support the first question as a main claim. The second question remains central. A monitoring-only sequence can represent $u_{t-K+1:t} \to r_{t+h}$, but it cannot encode which rock voxel is near which TBM component, whether that relation is geometrically compatible, or how the candidate relation set changes with chainage.

The graph representation is therefore justified by the structure of the domain, not by a preference for a particular learning architecture. TSP voxels, TBM surface samples, rock–rock adjacency, machine-surface adjacency, and rock–machine candidate relations are heterogeneous spatial entities and relations. The geometry-constrained graph makes these relations inspectable and auditable before any learning model is introduced.

6.2 What spatial interaction descriptors add beyond monitoring curves

Persistence is a strong description of short-horizon TBM response inertia. The revised method accepts this rather than competing against it directly. The role of the spatial graph is to describe the part of the excavation process that a monitoring curve cannot explicitly encode: component-specific exposure to nearby TSP-derived geological anomaly under geometry-compatible candidate relations.

The descriptors $A_c(t)$ and $I_c(t)$ provide a compact way to inspect this structure. $A_c(t)$ reports the pure geometric exposure of each TBM component to candidate rock–machine relations, while $I_c(t)$ reports the corresponding geology-weighted anomaly intensity. Comparing $I_c(t)$ with persistence residuals reframes the evaluation question from "does the graph beat persistence?" to "does the graph encode spatial diagnostic information consistent with the response changes left after persistence?" This is a more defensible question for the current data scale.

6.3 Interpretation boundary and future validation

The proposed descriptors should be interpreted as geometry-weighted summaries of TSP-derived anomaly around TBM components. They are not measured contact force, contact pressure, jamming probability, causal attribution, or calibrated operational risk. The current TSP anomaly score is intentionally simple, using a bounded low-velocity anomaly score derived from the training-reference distribution of V_p as the main formulation. Multi-attribute variants may be useful as sensitivity checks, but adding weights without independent validation would weaken the transparency of the method.

Future work should prioritise descriptor-specific residual tables, threshold sensitivity for active-zone and edge-construction parameters, alternative definitions of q_i, spatial detrending, multi-section validation, and independent engineering evidence such as event labels, contact observations, or expert assessments. Until such evidence is available, the defensible claim is a spatial interaction representation and response-residual diagnostic workflow, not recovery of true contact conditions or general predictive dominance.

7 Conclusion

This study develops a chainage-referenced spatial entity-relation framework for diagnostic analysis of rock–TBM interaction. The proposed framework formalises TSP-derived rock voxels, TBM surface components, and monitoring responses in a common excavation coordinate system, constructs geometry-constrained candidate rock–machine relations, and derives component–chainage spatial interaction descriptors from geometry-weighted TSP anomaly summaries.

The main contribution is not a claim of broad forecasting superiority. Persistence remains a strong temporal baseline, so the diagnostic target is the response change left after persistence rather than absolute response prediction. The defensible contribution is the explicit graph representation and the workflow that evaluates spatial interaction descriptors against persistence residuals of TBM responses. This framing preserves the need for graph representation while avoiding overclaims about contact-force recovery, jamming risk, or predictive dominance.

The current descriptor-level evidence shows the clearest support in the SJLS case and more limited, sample-sensitive support in BSLL. Future work should therefore focus on additional tunnel intervals, alternative but transparent definitions of the TSP-derived anomaly score, and independent engineering validation before the descriptors are connected to operational risk or actual contact conditions.

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Data availability statement

The raw TBM monitoring and TSP data are project data and are not publicly available because they contain engineering and site information. The processed experiment configurations, analysis scripts, and derived result summaries used to reproduce the reported tables and figures are available from the corresponding author upon reasonable request, subject to project data-use permissions.

References

[1] Danial Jahed Armaghani, Roohollah Shirani Faradonbeh Faradonbeh, Ehsan Momeni, Ahmad Fahimifar, and Mohd For Mohd Tahir. Performance prediction of tunnel boring machine through developing a gene expression programming equation. *Engineering with Computers*, 34(1):129–141, 2017. doi: 10.1007/s00366-017-0526-x.

[2] Stef De Sabbata and Pengyuan Liu. A graph neural network framework for spatial geodemographic classification. *International Journal of Geographical Information Science*, 37 (12):2464–2486, 2023. doi: 10.1080/13658816.2023.2254382.

[3] Xu Geng, Yaguang Li, Luheng Wang, Lingyu Zhang, Qiang Yang, Yan Liu, and Jieping Ye. Spatiotemporal multi-graph convolution network for ride-hailing demand forecasting. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33:3656–3663, 2019. doi: 10.1609/aaai.v33i01.33013656.

[4] Jafar Hassanpour, Jamal Rostami, and Jian Zhao. A new hard rock TBM performance prediction model for project planning. *Tunnelling and Underground Space Technology*, 26 (5):595–603, 2011. doi: 10.1016/j.tust.2011.04.004.

[5] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997. doi: 10.1162/neco.1997.9.8.1735.

[6] Shucai Li, Bin Liu, Xin Xu, Lichao Nie, Zhengyu Liu, Jie Song, Huaifeng Sun, Lei Chen, and Kairong Fan. An overview of ahead geological prospecting in tunneling. *Tunnelling and Underground Space Technology*, 63:69–94, 2017. doi: 10.1016/j.tust.2016.12.011.

[7] Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu. Diffusion convolutional recurrent neural network: data-driven traffic forecasting. *arXiv preprint arXiv:1707.01926*, 2018.

[8] Yang Liu, Qingsheng Guo, and Chuanbang Zheng. Urban street network morphology classification through street-block based graph neural networks and multi-model fusion. *International Journal of Digital Earth*, 18(1), 2025. doi: 10.1080/17538947.2025.2497490.

[9] Tobias Pfaff, Meire Fortunato, Alvaro Sanchez-Gonzalez, and Peter W. Battaglia. Learning mesh-based simulation with graph networks. *arXiv preprint arXiv:2010.03409*, 2021.

[10] Alvaro Sánchez-González, Jonathan Godwin, Tobias Pfaff, Rex Ying, Jure Leskovec, and Peter Battaglia. Learning to simulate complex physics with graph networks. *International Conference on Machine Learning*, pages 8459–8468, 2020.

[11] Michael Schlichtkrull, Thomas N. Kipf, Peter Bloem, Rianne van den Berg, Ivan Titov, and Max Welling. Modeling relational data with graph convolutional networks. *Lecture Notes in Computer Science*, 10843:593–607, 2018. doi: 10.1007/978-3-319-93417-4_38.

[12] X. Wu and A. T. Murray. A new approach to quantifying spatial contiguity using graph theory and spatial interaction. *International Journal of Geographical Information Science*, 22(4):387–407, 2008. doi: 10.1080/13658810701405615.

[13] Zonghan Wu, Shirui Pan, Guodong Long, Jing Chen, Chengqi Zhang, and S. Yu Philip. Connecting the dots: multivariate time series forecasting with graph neural networks. *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 753–763, 2020. doi: 10.1145/3394486.3403118.

[14] Saffet Yagiz. Utilizing rock mass properties for predicting TBM performance in hard rock condition. *Tunnelling and Underground Space Technology*, 23(3):326–339, 2008. doi: 10.1016/j.tust.2007.04.011.

[15] Chuxu Zhang, Dongjin Song, Chao Huang, Ananthram Swami, and Nitesh V. Chawla. Heterogeneous graph neural network. *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 793–803, 2019. doi: 10.1145/3292500.3330961.

[16] Jiani Zhang, Yu Zheng, and Dekang Qi. Deep spatio-temporal residual networks for city-wide crowd flows prediction. *Proceedings of the AAAI Conference on Artificial Intelligence*, 31(1):1655–1661, 2017.